

Design and Simulation of a Digital Access Card Verification System

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Introduction

- Digital access control regulates entry using electronic verification methods.
- Access cards store encoded credentials for secure authentication.
- Prevents unauthorized entry and protects sensitive areas & information.
- Digital logic circuits enable fast, reliable, and automated verification.

System Overview

Digital access control uses electronic verification to regulate entry.

- Access cards store encoded binary credentials for authentication
- Logic gates process card data to grant or deny access
- Combinational circuits validate card ID in real time
- Prevents unauthorized entry & protects sensitive areas
- Fast, reliable, and scalable digital verification



Problem Statement & Objectives

1. Traditional lock-and-key systems are vulnerable to theft, duplication, and unauthorized access. There is a growing need for a secure, efficient, and digital access control mechanism that can verify credentials reliably using combinational logic circuits.
2. Design a logic-based digital access card verification system using combinational circuits. Simulate the authentication process to validate correct card credentials. Demonstrate secure entry control through gate-level logic and binary encoding.

System Components

1.

Card Reader Module

Input interface that reads encoded card data. Converts card credentials into binary signals. Captures card ID via magnetic strip or RFID. Binary encoding ensures precise digital representation of access credentials.

2.

Verification Logic

Uses comparators to match input binary code with stored reference. AND/OR logic gates form the authentication circuit. Output signal grants or denies access based on gate evaluation results.

Methodology

Digital Logic Design Principles

- Boolean algebra used to define access rules
- Truth tables map all input combinations to valid/invalid outputs
- Combinational logic circuits process card data in real time
- Logic gates (AND, OR, NOT, XOR) form the core verification engine
- Circuit simulation performed using Logisim/Proteus
- Test vectors validate correct gate-level behavior

Design Approach

- Modular architecture: Input, Comparator, Decision, Output blocks
- Card ID encoded as 4-bit binary input signal
- Comparator module checks input against stored valid codes
- Combinational logic produces GRANT or DENY output
- LED indicators and buzzer simulate physical access response
- Design is scalable for multi-user and multi-door systems

Simulation Results

Access Granted: Valid card + PIN → Output HIGH (1)

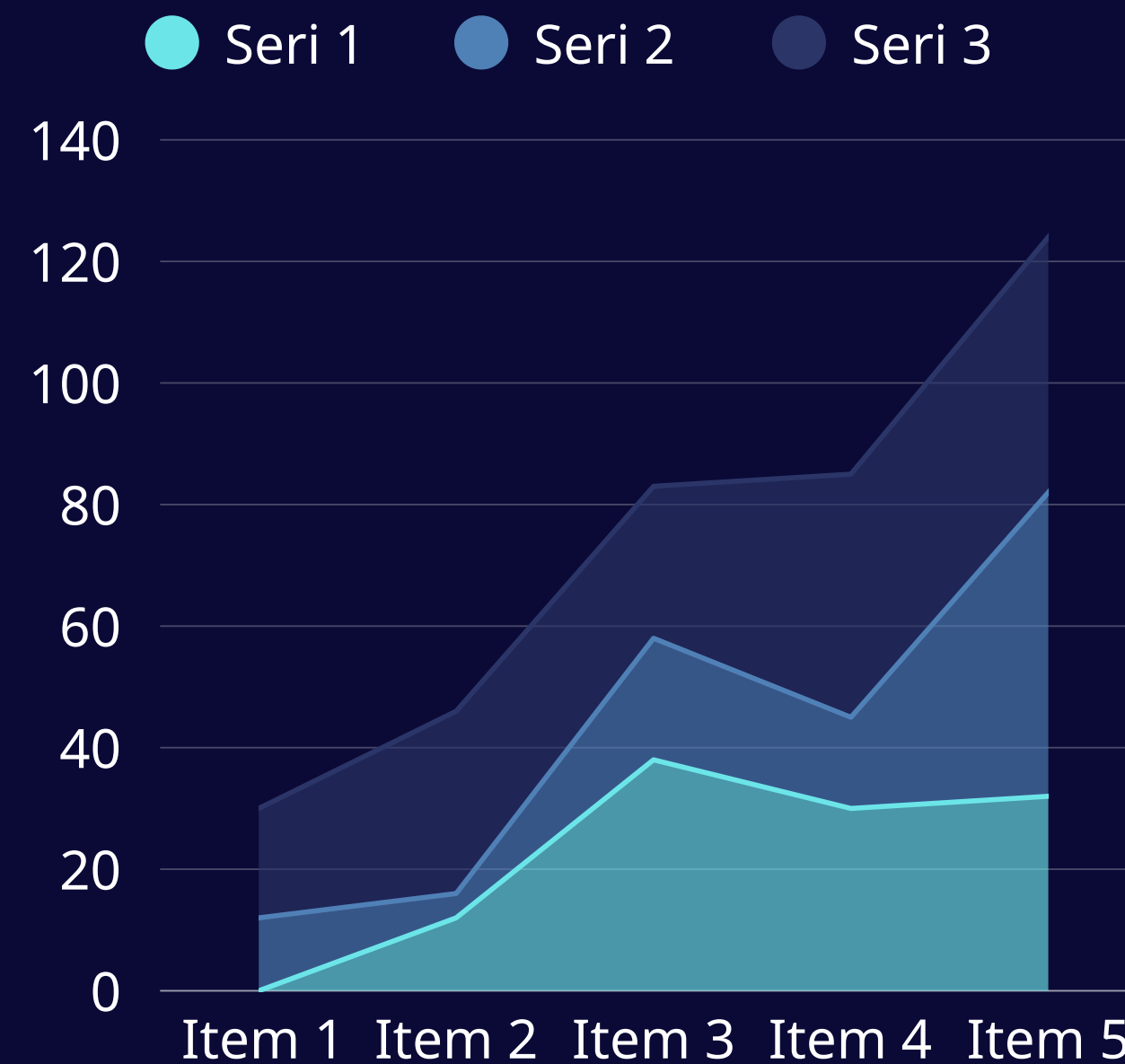
Access Denied: Invalid card or PIN → Output LOW (0)

Timing Analysis:

- Propagation Delay: ~15 ns per gate stage
- Total Verification Time: < 60 ns
- Clock Frequency: 10 MHz (stable)

Logic Outputs Verified:

- ✓ AND Gate: Correct HIGH only on full match
- ✓ NOT Gate: Inversion confirmed on mismatch
- ✓ XOR Gate: Detects bit-level discrepancies



Key Features & Advantages

Key Features:

- Fast Authentication – Instant card verification using combinational logic circuits
- Low Power Consumption – Efficient digital design minimizes energy usage
- Scalable Design – Easily expandable to support multiple access levels and users

Advantages:

- Enhanced Security – Multi-bit binary code verification prevents unauthorized access
- Reliable Operation – Logic gate-based system ensures consistent and error-free performance
- Easy Integration – Compatible with existing digital infrastructure and control systems

Applications & Future Scope

Applications:

- Building Security – Restrict entry to authorized personnel only
- Lab & Server Room Access – Protect sensitive equipment and data
- Restricted Area Control – Secure zones in offices, hospitals, campuses

Future Scope:

- RFID Integration – Contactless card reading for faster verification
- Biometric Enhancement – Fingerprint/face recognition as second factor
- IoT Connectivity – Remote monitoring and real-time access logging
- Cloud-Based Management – Centralized control across multiple locations

Conclusion

Successfully designed and simulated a Digital Access Card Verification System using combinational logic circuits. The system demonstrates secure, efficient, and reliable authentication through logic gate implementation. Binary-encoded card data is validated accurately, preventing unauthorized access. The design proves that digital logic can form the backbone of real-world access control systems — scalable, cost-effective, and robust.



Thank You!
Questions &
Discussion